The complete Linux filesystem guide.

14 DIRECTORIES

/etc	/var	/home	/tmp	/proc	/sys	/usr
/root	/opt	/dev	/boot	/bin	/lib	/mnt

each one explained →

```
$ ls /
# 14 paths every Linux user touches.
```

/etc

System configuration files

```
root@localhost: ~
root@localhost:~# ls /etc
passwd  shadow  sudoers  hosts
crontab  ssh    nginx   resolv.conf
root@localhost:~# cat /etc/passwd | head -1
root:x:0:0:root:/root:/bin/bash
```

FILES

passwd	User accounts (UID, GID, shell)
shadow	Hashed user passwords (root only)
sudoers	Privilege escalation rules
crontab	System-wide scheduled jobs

OVERVIEW

/etc holds plain-text configuration for the system and most installed services. Editing here changes behavior globally → always back up and validate (visudo, sshd -t) before reload.

/var

Variable, ever-changing data

```
root@localhost: ~
root@localhost:~# ls /var
log     mail    spool    cache
lib     tmp     www      backups
root@localhost:~# tail -1 /var/log/auth.log
sshd[1234]: Accepted publickey for kerem
```

FILES

log/	System and application logs
mail/	Mail queues and mailboxes
spool/	Print, mail, and cron queues
cache/	Application cache data

OVERVIEW

/var holds files whose contents change as the system runs - logs, queues, caches, and databases. Logs in /var/log are critical for debugging and incident response → check first.

/home

User home directories

```
root@localhost: ~
root@localhost:~# ls /home
kerem  alice  bob
root@localhost:~# ls -la ~kerem
.bashrc  .ssh  .config  Documents
```

FILES

.bashrc	Per-user shell configuration
.ssh/	SSH keys, authorized_keys, config
.gnupg/	GPG keys and trust database
Documents/	User personal data

OVERVIEW

/home contains a separate directory for each regular user, owned by that user. Dotfiles store per-user config; .ssh and .gnupg hold credentials → keep them mode 700 / 600.

/tmp

World-writable temporary files

```
root@localhost: ~
root@localhost:~# ls -ld /tmp
drwxrwxrwt 17 root root 4096 Apr 29 /tmp
root@localhost:~# ls /tmp
systemd-private-*  sess_*  npm-*
```

FILES

perms	1777: world-writable + sticky bit
sticky bit	Only owner can delete their files
/tmp clean	Cleared on every reboot
watch list	Common malware drop zone

OVERVIEW

/tmp is world-writable scratch space cleared on reboot. The sticky bit (1777) lets users write but only owners delete their files. Watch attackers staging payloads here.

/proc

Live kernel & process state (virtual)

```
root@localhost: ~
root@localhost:~# ls /proc | head -3
1  2  3
root@localhost:~# cat /proc/cpuinfo | head -2
processor : 0
vendor_id : GenuineIntel
```

FILES

cpuinfo	CPU vendor, model, flags, cores
meminfo	Memory total, free, swap, cached
mounts	Currently mounted filesystems
[PID]/	Per-process kernel data

OVERVIEW

/proc is a virtual filesystem the kernel generates on the fly. Numbered dirs (PIDs) expose every running process; files like cpuinfo and meminfo reveal live system state.

/sys

Kernel and device interface (sysfs)

```
root@localhost: ~
root@localhost:~# ls /sys
block  bus  class  devices  fs  kernel  module
root@localhost:~# cat /sys/class/net/eth0/address
02:42:ac:11:00:02
```

FILES

class/	Devices grouped by category
devices/	Hardware tree (PCI, USB, etc.)
kernel/	Kernel internals and parameters
module/	Currently loaded kernel modules

OVERVIEW

/sys is the modern interface for kernel and hardware data, replacing many older /proc files. Reading it reflects device state; writing to certain files tunes the kernel.

/usr

User programs and resources

```
root@localhost: ~

root@localhost:~# ls /usr
bin  lib  share  local  include
root@localhost:~# which python3
/usr/bin/python3
```

FILES

bin/	Most user binaries (commands)
lib/	Libraries used by /usr/bin
share/	Architecture-independent data
local/	Manually installed software

OVERVIEW

/usr is the secondary hierarchy where most installed software lives. /usr is read-only on hardened systems. /usr/local is the place for software you compile or install yourself.

/root

Root user's home directory

root@localhost: ~

```
root@localhost:~# ls -ld /root
drwx------ 5 root root 4096 Apr 29 /root
root@localhost:~# sudo ls /root
.bash_history  .ssh  scripts
```

FILES

vs /	/root ≠ / (filesystem root)
permissions	Mode 700: only root can read
.bash_history	Root command log (audit gold)
admin scripts	Often live here on hardened hosts

OVERVIEW

/root is the home directory of the root account, separate from / (filesystem root). Mode 700 keeps regular users out. Audit .bash_history and any cron jobs you find.

/opt

Optional / third-party software

```
root@localhost: ~

root@localhost:~# ls /opt
google  jetbrains  lampp
root@localhost:~# ls /opt/google/chrome
chrome  chrome.png  resources
```

FILES

google/	Chrome and other Google software
jetbrains/	IntelliJ, PyCharm, GoLand, etc.
self-contained	Each vendor isolated
vs /usr	Outside the package manager

OVERVIEW

/opt is for optional, self-contained third party packages installed outside the system package manager. Each vendor gets its own subdirectory and ships its own libs and bins.

/dev

Device files

```
root@localhost: ~
root@localhost:~# ls /dev | head -3
null  zero  urandom
root@localhost:~# head -c 16 /dev/urandom | xxd
00000000: a1b2 c3d4 e5f6 0789 1a2b 3c4d 5e6f 7081
```

FILES

null	Discards anything written to it
zero	Endless stream of null bytes
urandom	Cryptographic random source
sda, sdb	Block devices: physical disks

OVERVIEW

/dev contains special files the kernel exposes as devices. Reading or writing them talks to hardware or kernel features. Numbers (sda1) refer to disk partitions.

/boot

Kernel and bootloader files

```
root@localhost:~# ls /boot
vmlinuz-6.5  initrd.img-6.5  grub  config-6.5
root@localhost:~# uname -r
6.5.0-21-generic
```

FILES

vmlinuz-*	Compressed Linux kernel
initrd.img-*	Initial ramdisk for early boot
grub/	GRUB bootloader configuration
config-*	Kernel build configuration

OVERVIEW

/boot holds files needed to boot the system: the kernel, initial ramdisk, and bootloader. Removing files here can leave your system unbootable → touch with extreme care.

/bin

Essential system binaries

```
root@localhost: ~
root@localhost:~# ls /bin | head -3
bash  cat   ls
root@localhost:~# readlink /bin
usr/bin
```

FILES

bash, sh	Shells: command interpreters
ls, cat	List and read files
cp, mv, rm	Copy, move, remove files
symlink	On modern distros → /usr/bin

OVERVIEW

/bin contains the most essential commands available to all users. On modern systems /bin is a symlink to /usr/bin (merged-usr), unifying the binary directories.

/lib

Essential shared libraries

root@localhost: ~

```
root@localhost:~# ls /lib | head -3
ld-linux-x86-64.so.2  libc.so.6  modules
root@localhost:~# ldd /bin/bash | head -1
linux-vdso.so.1 (0x00007ffd5c3a1000)
```

FILES

libc.so.6	C standard library (glibc)
ld-linux-*	Dynamic linker / loader
modules/	Loadable kernel modules
***.so files**	Shared object libraries

OVERVIEW

/lib holds libraries needed by binaries in /bin and /sbin. ld-linux loads them at startup. /lib/modules holds loadable kernel modules → a frequent rootkit target.

/mnt

Manual mount points

```
root@localhost: ~

root@localhost:~# ls /mnt
usb  backup  share
root@localhost:~# mount | grep /mnt
/dev/sdb1 on /mnt/usb type ext4 (rw)
```

FILES

usb/	Manual external device mount
backup/	Network share or extra disk
custom dirs	Anything you mount yourself
vs /media	/media is for auto-mounts

OVERVIEW

/mnt is the conventional place to mount filesystems by hand (USB drives, network shares). /media is reserved for automatic, system-managed mounts → use the right one.

THANKS FOR READING

Save this guide.

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